Connected Fiber Energy in a Primitive Möbius Tail: Ramanujan-Mean Closure, Divisor-Lattice Reassembly, and a Shared-Factor Character Gate

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Abstract

TPC-20 compresses the compatible short–short part of a primitive Möbius residual correlation into one finite Fourier transform on each unit fiber modulo q. Its exact mean–discrepancy split leaves two apparently separate analytic obligations. We close the mean obligation and replace the remaining arbitrary-fiber formulation by two exact normal forms that retain the row provenance.

Put $H = \operatorname{rad}|h|$, $QJ \asymp X$, and let $F \in C_c^\infty(\mathbb{R})$. The Ramanujan mean packet at modulus q is exactly

$$\overline{Z}_q \left\{ \sum_{(j,Hq)=1} F(j/J) - J\frac{\varphi(Hq)}{Hq}\widehat{F}(0) \right\}.$$

The expression in braces is $O_F(\tau(Hq))$, uniformly even when $Hq \asymp J$. Under the standard separated-packet bound $\mathcal{Z}_1 \ll Q^2X^\varepsilon$, absolute summation over every short–short divisor pair gives $O_\varepsilon(Q^2X^\varepsilon)$; this scale is $\asymp XQJ^{-1}X^\varepsilon$. Hence throughout the strict single-leg range of TPC-20 the complete Ramanujan mean has a fixed power saving and is negligible with arbitrary logarithmic saving.

The actual row geometry also gives the unconditional occupancy bound

$$\|D_q\|_{2,\mathcal{U}_q} \ll_\varepsilon Q^2q^{-1/2}X^\varepsilon.$$

On $q \asymp Y$, however, its insertion into the restricted-frequency estimate saves only when $Y < J$ by a fixed power. Thus occupancy alone recovers the already-completable joint-period range and does not enter the genuinely long regime.

For a rank-one row packet, lcm aggregation is a divisor-lattice Möbius difference of cumulative row-fiber products. For the actual generic Schur mask, multiplicative Fourier inversion gives a sharper normal form: the nonprincipal fiber energy is exactly a shared-g convolution of masked two-row Dirichlet-character sums, where $g = (u, v)$. In the unmasked case these sums factor into one-row Dirichlet polynomials; in the generic case the determinant and gcd mask remains inside a connected fourth moment.

We finally prove that scalar degenerate removal cannot be moved through the nonlinear fiber L^2 norm merely by citing its scalar saving. A single row diagonal produces exactly the one-fiber spike of TPC-20 and saturates every nonprincipal multiplicative frequency relative to its coefficient amplitude, although its complete scalar correlation is power-saving. The correct next input is therefore a masked shared-factor character fourth-moment estimate, not an unpruned Gram norm. We do not prove the short–short discrepancy estimate, control the large-divisor legs, breach sieve parity, or obtain a fixed-shift prime-pair asymptotic.

Keywords: Möbius tail; CRT fibers; Ramanujan sums; divisor-lattice inversion; Dirichlet characters; connected fourth moments; sieve parity.

MSC 2020: Primary 11N35, 11N37; Secondary 11L05, 11L07, 11L40.

1 Introduction

The symmetric Möbius tail isolated in TPC-17 is reduced in TPC-18 to a primitive two-row residual correlation. After diagonal, same-source, near-determinant, and large-gcd pairs are removed by elementary counting, a failed block forces a generic correlation at the natural scale XQ [5]. TPC-19 gives exact divisor and Poisson normal forms for that correlation, but its four locally centered channels cannot be estimated independently without losing their matching [6]. TPC-20 performs the matching first. In the strict single-leg range it completes the two one-divisor spectra and compresses the remaining compatible short–short joint spectrum into one ordinary finite Fourier transform on $\mathcal{U}_q = (\mathbb{Z}/q\mathbb{Z})^\times$ [7].

The last compression has the exact split

$$S_q(r) = \overline{Z}_q\, c_q(r) + \widehat{D}_q(r), \qquad \sum_{r \bmod q} |\widehat{D}_q(r)|^2 = q \sum_{\kappa\in\mathcal{U}_q} |D_q(\kappa)|^2. \tag{1}$$

TPC-20 deliberately leaves both terms on the right as analytic inputs. The purpose of the present paper is to decide which part is elementary, which part has additional row structure, and which part is still a new arithmetic estimate.

1.1 The mean packet closes absolutely

Our first result removes one of the two inputs. For $M \geq 1$, define

$$\Delta_M(F;J) = \frac{J}{M}\sum_{r\neq 0} c_M(r)\widehat{F}\left(\frac{rJ}{M}\right). \tag{2}$$

Reduced-class Poisson summation and ordinary Möbius inversion give

$$\Delta_M(F;J) = \sum_{(j,M)=1} F(j/J) - J\frac{\varphi(M)}{M}\widehat{F}(0), \qquad |\Delta_M(F;J)| \ll_F \tau(M). \tag{3}$$

The estimate is uniform through the transition $M \asymp J$; it is a bounded smooth lattice discrepancy, not a long-modulus prime theorem. The TPC-20 mean packet is exactly $\overline{Z}_q\Delta_{Hq}(F;J)$. Under the standard separated-packet bound $\mathcal{Z}_1 \ll Q^2X^\varepsilon$, summing it absolutely over all short–short lcm layers gives

$$\sum_q |\mathcal{L}_q^{\mathrm{mean}}| \ll_\varepsilon Q^2X^\varepsilon, \qquad Q^2X^\varepsilon \asymp \frac{XQ}{J}X^\varepsilon. \tag{4}$$

Thus the mean is power-saving throughout the strict single-leg range. No cancellation between moduli is used.

1.2 What row occupancy proves, and where it stops

The fiber vector is not completely arbitrary. Because $u \leq R < L/2$, fixing d and a row class modulo u confines the prime source ℓ to one progression containing $O(L/u)$ integers. Consequently each fiber contains at most $O(Q/u)$ rows on the u-side. This gives the unconditional estimate

$$\|D_q\|_{2,\mathcal{U}_q} \ll_\varepsilon \frac{Q^2}{\sqrt{q}}X^\varepsilon. \tag{5}$$

For $q \asymp Y$, insertion into the TPC-20 restricted-frequency inequality produces

$$\sum_{q \asymp Y} |\mathcal{L}_q^{\mathrm{disc}}| \ll_\varepsilon Q^2 \sqrt{JY}\, X^\varepsilon. \tag{6}$$

Relative to $XQ = Q^2 J$, this saves a fixed power only when $Y < J$ by a fixed power. That is precisely the region where the whole joint period is already short. Occupancy improves the abstract one-fiber bound, but it does not cross the long-lcm boundary.

1.3 Two provenance-preserving normal forms

The original TT^* row weights have rank-one provenance before the generic Schur mask is imposed. For a separated bilinear packet, let $A_u^x(\kappa)$ and $A_u^y(\kappa)$ be the two one-row sums in the class $m\kappa \equiv -h \pmod u$, and put

$$B_s^x(\kappa) = \sum_{u|s} \vartheta_R(u) A_u^x(\kappa), \qquad B_s^y(\kappa) = \sum_{u|s} \vartheta_R(u) A_u^y(\kappa). \tag{7}$$

The unmasked lcm layer has the exact divisor-lattice formula

$$Z_q^{x,y}(\kappa) = \sum_{s|q} \mu(q/s) B_s^x(\kappa \bmod s) B_s^y(\kappa \bmod s). \tag{8}$$

In the Hermitian case this is a Möbius difference of cumulative nonnegative energies. CRT lifting then shows that on frequencies coprime to q, the outer lcm Möbius signs all disappear. This is structure, but not automatic cancellation.

For the actual generic mask, multiplicative Fourier analysis is more faithful. Write $q = gab$, $u = ga$, $v = gb$, with g, a, b pairwise coprime. If $\chi = \chi_g \chi_a \chi_b$ is a character of $\mathcal{U}_q$, the fiber transform is exactly a convolution over characters ψ of the shared factor g:

$$\widehat{Z}_{u,v}^{\mathfrak{m}}(\chi) = \frac{\overline{\chi(-h)}}{\varphi(g)} \sum_{\psi \bmod g} \mathcal{B}_{\mathfrak{m}}(\chi_g \chi_a \psi, \chi_b \overline{\psi}). \tag{9}$$

Here $\mathfrak{m}(m_1, m_2)$ is the generic Schur mask and $\mathcal{B}_{\mathfrak{m}}$ is its masked two-row Dirichlet-character sum. When $\mathfrak{m} \equiv 1$, the latter factors into two one-row Dirichlet polynomials. For the generic mask it does not. Multiplicative Parseval therefore identifies the exact surviving input as a masked shared-factor character fourth moment.

1.4 Why scalar unpruning cannot be put inside the norm

The isolated short–short signal is $O_\varepsilon(X^\varepsilon)$ pointwise. Hence the TPC-18 tuple counts allow the degenerate row pairs to be restored and removed at the level of the complete scalar correlation, with fixed-power error. It is tempting to restore the full Gram packet, take its fiber L^2 norm, and then subtract the degenerate scalar terms. This is invalid because the norm is nonlinear.

We make the failure exact. A single row diagonal contributes to one class κ_q only:

$$Z_{q,m}^{\mathrm{rowdiag}}(\kappa) = |x_m|^2 \Gamma_\vartheta(q) \mathbf{1}_{\{\kappa = \kappa_q\}}, \tag{10}$$

and its multiplicative Fourier magnitude is constant in every nonprincipal frequency. Thus a row diagonal can saturate the one-fiber obstruction of TPC-20 even though its summed physical correlation is already negligible. The generic mask must remain inside the connected character moment, or the degenerate pieces must be removed before a nonlinear norm is applied.

The paper therefore makes a positive reduction without claiming a parity breakthrough: the Ramanujan mean is closed, the arbitrary fiber is replaced by an exact masked character object, and two insufficient shortcuts are ruled out quantitatively. The masked fourth moment and the large-divisor legs remain open.

1.5 Organization

Section 2 fixes the inherited packet and its generic mask. Section 3 closes the Ramanujan mean. Section 4 proves the occupancy frontier. Section 5 gives the divisor-lattice reassembly and its additive Fourier lift. Section 6 derives the exact shared-factor character gate. Section 7 proves the diagonal-spike obstruction and the scalar/norm distinction. Section 8 audits the remaining analytic interface, and Section 9 records the exact finite certificate and the conclusion.

2 The inherited primitive fiber interface

Fix $h \in \mathbb{Z} \setminus \{0\}$, put $H = \operatorname{rad}|h|$, and let

$$e(t) = \exp(2\pi i t), \qquad e_s(t) = e(t/s).$$

$$R = \lfloor X^{1/2-\delta} \rfloor, \qquad V = \lfloor R^{1/2} \rfloor, \qquad 0 < \delta < \frac{1}{2}. \tag{11}$$

Take X large enough that $R \geq H^2$. Then the inherited restriction $d \leq V$ implies $dH \leq R$, as required in the TPC-20 recombination. The row parameters satisfy

$$Q = LD, \qquad QJ \asymp X, \qquad L > \max(2R, 2|h|), \tag{12}$$

and

$$\ell \in [L/2, 2L] \text{ is prime}, \qquad d \in [D, 2D] \cap [1, V], \qquad m = \ell d. \tag{13}$$

The integer d is squarefree, $(d, H) = 1$, and the orbit is restricted to $(j, H) = 1$. In particular $(m, H) = 1$. All implied constants may depend on h, δ, and fixed smooth weights.

TPC-20 supplies two short–short coefficient choices,

$$\vartheta_R(u) = \lambda'_R(u) \quad \text{or} \quad \vartheta_R(u) = b_R(u)\mathbf{1}_{u \leq R}, \tag{14}$$

where both are supported on squarefree $u \leq R$ and, for every fixed $\varepsilon > 0$,

$$|\vartheta_R(u)| \ll_\varepsilon X^\varepsilon. \tag{15}$$

Both coefficient choices are real-valued. This fact is used whenever a Hermitian packet is written as an absolute square. The first choice is the centered finite Ramanujan model. The second is the matched residual coefficient restricted to its two short divisor legs. No statement below controls the complementary terms with $u > R$ or $v > R$.

2.1 Rows, masks, and fiber vectors

Let $\mathcal{A}$ be one finite separated row packet. Its labels $\alpha = (\ell, d)$ satisfy (13), and its weighted cardinality obeys

$$\sum_{\alpha \in \mathcal{A}} |x_\alpha| + \sum_{\alpha \in \mathcal{A}} |y_\alpha| \ll QX^\varepsilon \tag{16}$$

for the bounded row weights used below. The original smooth factors in mj/X and dj/K admit the standard rapidly convergent Mellin–Fourier separation. Each separated term has a common $F(j/J)$ and rank-one row weights $x_{\alpha_1} y_{\alpha_2}$; the total projective mass of the truncated superposition is X^ε, with arbitrary power error.

The generic TPC-18 row-pair mask is

$$\mathfrak{m}(\alpha_1, \alpha_2) = \mathbf{1}_{\{\ell_1 \neq \ell_2\}} \mathbf{1}_{\{|m_1 - m_2| > QX^{-\kappa}\}} \mathbf{1}_{\{(d_1, d_2) \leq X^\kappa\}}, \tag{17}$$

where $\kappa > 0$ is fixed. We also allow an arbitrary fixed row-pair restriction independent of j, denoted by the same letter, when proving the exact fiber identities.

For squarefree $u, v \le R$, put

$$g = (u, v), \qquad q = [u, v]. \tag{18}$$

All aggregated layers used below are restricted to $(q, H) = 1$. Layers with $(q, H) > 1$ have empty admissible fibers and are omitted; in particular $\overline{H} \pmod q$ is always defined. If

$$(u, m_1 H) = (v, m_2 H) = 1,$$

then the pair (m_1, m_2) is compatible precisely when $g \mid m_1 - m_2$. It then determines the unique unit class

$$\kappa \in \mathcal{U}_q = (\mathbb{Z}/q\mathbb{Z})^\times, \qquad m_1\kappa \equiv -h \pmod u, \quad m_2\kappa \equiv -h \pmod v. \tag{19}$$

If either row–modulus admissibility condition fails, the corresponding fiber is zero. Define

$$Z_{u,v}^{\mathfrak{m}}(\kappa) = \sum_{\alpha_1,\alpha_2 \in \mathcal{A}} x_{\alpha_1} y_{\alpha_2} \mathfrak{m}(\alpha_1, \alpha_2) \mathbf{1}_{\{m_1 \equiv -h\overline{\kappa}\ (u)\}} \mathbf{1}_{\{m_2 \equiv -h\overline{\kappa}\ (v)\}}, \tag{20}$$

$$Z_q^{\mathfrak{m}}(\kappa) = \sum_{\substack{u,v \le R \\ [u,v]=q}} \vartheta_R(u)\vartheta_R(v) Z_{u,v}^{\mathfrak{m}}(\kappa). \tag{21}$$

Congruences modulo 1 are vacuous. An inadmissible divisor–row combination contributes zero. In the retained low nonzero joint frequencies of TPC-20, the common multiplier is g, and the factor $g/(uv) = 1/q$ gives the scalar packet

$$\mathcal{L}_q = \frac{J}{Hq} \sum_{r \ne 0} c_H(r) \widehat{F}\left(\frac{rJ}{Hq}\right) S_q(r), \qquad S_q(r) = \sum_{\kappa \in \mathcal{U}_q} Z_q^{\mathfrak{m}}(\kappa) e_q(r\overline{H}\kappa). \tag{22}$$

Put

$$\overline{Z}_q^{\mathfrak{m}} = \frac{1}{\varphi(q)} \sum_{\kappa \in \mathcal{U}_q} Z_q^{\mathfrak{m}}(\kappa), \qquad D_q^{\mathfrak{m}}(\kappa) = Z_q^{\mathfrak{m}}(\kappa) - \overline{Z}_q^{\mathfrak{m}}. \tag{23}$$

Then

$$S_q(r) = \overline{Z}_q^{\mathfrak{m}} c_q(r) + \widehat{D}_q^{\mathfrak{m}}(r), \qquad \sum_{r \bmod q} |\widehat{D}_q^{\mathfrak{m}}(r)|^2 = q \|D_q^{\mathfrak{m}}\|_{2,\mathcal{U}_q}^2. \tag{24}$$

The mean term is closed in section 3; the discrepancy is the subject of the remaining sections.

2.2 The inherited restricted-frequency gate

For a long-modulus family $\mathcal{Q}$, TPC-20 proves, after truncating the rapidly decaying Fourier tail, that

$$\sum_{q \in \mathcal{Q}} |\mathcal{L}_q^{\mathrm{disc}}| \ll_{H,F,\varepsilon} X^{\varepsilon/2} \sqrt{\frac{J}{H}} \sum_{q \in \mathcal{Q}} \|D_q^{\mathfrak{m}}\|_{2,\mathcal{U}_q} + O_A(X^{-A}\mathcal{Z}_1), \tag{25}$$

under its no-wrap and polynomial-mass hypotheses. Here $\mathcal{Z}_1$ denotes the relevant total coefficient mass. The natural row-pair–orbit scale is

$$Q^2 J \asymp XQ. \tag{26}$$

Every use of (25) below is compared with this scale. The estimate is sufficient, not known to be necessary; in particular, scalar cancellation may be invisible to the nonlinear fiber norm.

3 The Ramanujan mean as a coprime-lattice discrepancy

We first dispose of the deterministic mean packet left by the CRT-fiber decomposition of TPC-20. The argument applies to one separated compatible short–short packet. Thus the same function $F(j/J)$, with F in a uniformly bounded subset of $C_c^\infty(\mathbb{R})$, occurs for every row pair, and all separated row weights are absorbed into coefficients z_{m_1,m_2}. We use the Fourier convention

$$\widehat{F}(\xi) = \int_{\mathbb{R}} F(x)e(-x\xi)\,dx. \tag{27}$$

The divisor coefficient ϑ_R denotes either

$$\vartheta_R(u) = \lambda'_R(u) \qquad \text{or} \qquad \vartheta_R(u) = b_R(u)\mathbf{1}_{\{u\le R\}}, \tag{28}$$

and is extended by zero outside its squarefree support.

For admissible $u, v \le R$, define the compatible row-pair mass

$$\mathfrak{R}_{u,v} := \sum_{(m_1,m_2)} z_{m_1,m_2}\mathbf{1}_{\{(u,m_1H)=1\}}\mathbf{1}_{\{(v,m_2H)=1\}}\mathbf{1}_{\{(u,v)|m_1-m_2\}}. \tag{29}$$

Any row-pair restriction independent of the orbit variable may be included in this sum. The compatible-fiber bijection implies

$$\sum_{\kappa\in(\mathbb{Z}/[u,v]\mathbb{Z})^\times} Z_{u,v}(\kappa) = \mathfrak{R}_{u,v}. \tag{30}$$

Consequently, after aggregating all pairs with the same least common multiple,

$$\boxed{\overline{Z}_q = \frac{1}{\varphi(q)} \sum_{\substack{u,v\le R\\ [u,v]=q}} \vartheta_R(u)\vartheta_R(v)\mathfrak{R}_{u,v}.} \tag{31}$$

Every nonzero term has $(q, H) = 1$. No equidistribution of the row weights among the unit fibers is used in (31).

3.1 Exact reduced-class Poisson inversion

For an integer $M \ge 1$, put

$$\Delta_M(F; J) = \frac{J}{M} \sum_{r\in\mathbb{Z}\setminus\{0\}} c_M(r)\widehat{F}\left(\frac{rJ}{M}\right). \tag{32}$$

Proposition 3.1 (The Ramanujan spectrum is a coprime-lattice error)**.** *For every $M \ge 1$ and $J > 0$, one has the exact identity*

$$\boxed{\Delta_M(F; J) = \sum_{\substack{n\in\mathbb{Z}\\ (n,M)=1}} F(n/J) - \frac{\varphi(M)}{M} J\widehat{F}(0).} \tag{33}$$

Moreover, uniformly in M and J,

$$|\Delta_M(F; J)| \ll_F \tau(M). \tag{34}$$

The implied constant is uniform when F ranges over a uniformly bounded subset of $C_c^\infty(\mathbb{R})$.

Proof. For each reduced class a mod M, Poisson summation on the progression $a + M\mathbb{Z}$ gives

$$\sum_{k\in\mathbb{Z}} F\left(\frac{a+kM}{J}\right) = \frac{J}{M}\sum_{r\in\mathbb{Z}} e_M(ra)\widehat{F}\left(\frac{rJ}{M}\right).$$

Summing over a mod M with $(a, M) = 1$ produces $c_M(r)$. The term $r = 0$ is $J\varphi(M)\widehat{F}(0)/M$; removing it proves (33).

For the uniform bound, insert $\mathbf{1}_{\{(n,M)=1\}} = \sum_{d|(n,M)} \mu(d)$ in the physical-space side. This gives

$$\Delta_M(F; J) = \sum_{d|M} \mu(d) \left\{ \sum_{k\in\mathbb{Z}} F(dk/J) - \frac{J}{d}\widehat{F}(0) \right\}. \tag{35}$$

The expression in braces is $O_F(1)$, uniformly for $d, J > 0$. Indeed, when $J/d \geq 1$, the standard bounded-variation estimate for a Riemann sum gives this bound. When $J/d < 1$, compact support shows that the discrete sum contains $O_F(1)$ points, while $(J/d)\widehat{F}(0) = O_F(1)$. Summing over $d \mid M$ proves (34). □

The deterministic part of the TPC-20 long packet was

$$\mathcal{L}_q^{\mathrm{mean}} = \overline{Z}_q \frac{J}{Hq} \sum_{r\neq 0} c_{Hq}(r)\widehat{F}\left(\frac{rJ}{Hq}\right).$$

Thus proposition 3.1 gives the exact physical-space form

$$\boxed{\mathcal{L}_q^{\mathrm{mean}} = \overline{Z}_q\, \Delta_{Hq}(F; J).} \tag{36}$$

This identity concerns only the chosen compatible short–short packet; it does not include either large-divisor leg of the full residual.

3.2 The arithmetic cost of all least common multiples

Lemma 3.2 (Summable lcm cost)**.** *Assume $R \leq X$ and let ϑ_R be either coefficient in (28). For every $\varepsilon > 0$,*

$$\boxed{\sum_{u,v\leq R} \frac{|\vartheta_R(u)\vartheta_R(v)|\tau(H[u,v])}{\varphi([u,v])} \ll_{\varepsilon,H} X^{\varepsilon}.} \tag{37}$$

Proof. The divisor formula for the finite model gives, on squarefree support,

$$|\lambda_R'(u)| \leq \frac{u}{\varphi(u)} \sum_{b\leq R/u} \frac{\mu(b)^2}{\varphi(b)} \ll_\eta R^\eta$$

for every $\eta > 0$. Here one may use $n/\varphi(n) \ll_\eta n^\eta$ and $1/\varphi(b) \ll_\eta b^{-1+\eta}$. Since $b_R(u) = -\mu(u)\log u - \lambda_R'(u)$, the same bound, after adjusting η, holds for both choices of ϑ_R.

Put $q = [u, v]$. The standard divisor estimates imply

$$\frac{\tau(Hq)}{\varphi(q)} = \frac{1}{q}\tau(Hq)\frac{q}{\varphi(q)} \ll_{\eta,H} \frac{X^\eta}{q},$$

because $q \leq R^2 \leq X^2$. It remains to note that

$$\begin{aligned}\sum_{u,v\leq R} \frac{1}{[u,v]} &\leq \sum_{g\leq R} \frac{1}{g} \left(\sum_{a\leq R/g} \frac{1}{a}\right)\left(\sum_{b\leq R/g} \frac{1}{b}\right) \\ &\ll \log^3(2R).\end{aligned} \tag{38}$$

Indeed, write $u = ga$, $v = gb$ with $g = (u, v)$, and then drop the restriction $(a, b) = 1$. Combining these estimates and choosing η as a sufficiently small fixed fraction of ε proves (37). □

3.3 Absolute closure of the mean packets

Let

$$\mathcal{Z}_1 = \sum_{(m_1,m_2)} |z_{m_1,m_2}| \tag{39}$$

for the separated row packet under consideration.

Theorem 3.3 (Summed Ramanujan-mean closure)**.** *Let $\mathcal{Q}$ be any collection of denominators arising from the chosen compatible short–short packet. For either coefficient in (28),*

$$\boxed{\sum_{q\in\mathcal{Q}} |\mathcal{L}_q^{\mathrm{mean}}| \ll_{\varepsilon,F,H} \mathcal{Z}_1 X^{\varepsilon}.} \tag{40}$$

In particular, if the TPC row packet satisfies

$$\mathcal{Z}_1 \ll Q^2 X^{\varepsilon}, \tag{41}$$

then

$$\sum_{q\in\mathcal{Q}} |\mathcal{L}_q^{\mathrm{mean}}| \ll_{\varepsilon,F,H} Q^2 X^{\varepsilon}. \tag{42}$$

Proof. By (36), (34), and (31),

$$\sum_{q\in\mathcal{Q}} |\mathcal{L}_q^{\mathrm{mean}}| \ll_F \sum_{q\in\mathcal{Q}} \frac{\tau(Hq)}{\varphi(q)} \sum_{\substack{u,v\le R\\ [u,v]=q}} |\vartheta_R(u)\vartheta_R(v)| \sum_{(m_1,m_2)} |z_{m_1,m_2}|.$$

Dropping the compatibility and row-admissibility indicators only increases this absolute majorant. We may therefore interchange the row and divisor sums and apply lemma 3.2. This proves (40); (42) follows from (41) after renaming ε. ☐

Corollary 3.4 (Fixed-power saving in the strict single-leg range)**.** *Assume*

$$QJ \asymp X, \qquad Q \le X^{1/2+\delta-\eta_0}, \qquad 0<\delta<\frac{1}{2}, \qquad \eta_0>0, \tag{43}$$

and (41). Then for every $B>0$,

$$\boxed{\sum_{q\in\mathcal{Q}} |\mathcal{L}_q^{\mathrm{mean}}| \ll_{B,F,H,\delta,\eta_0} XQ(\log X)^{-B}.} \tag{44}$$

More precisely, the ratio of (42) to the natural row-pair–orbit scale $XQ = Q^2J$ is $O(X^{\varepsilon}/J)$, while

$$J \gg X^{1/2-\delta+\eta_0}. \tag{45}$$

For the largest divisor slice in the two profiles recorded in TPC-20, the corresponding orbit exponents are respectively

$$J \ge X^{23/84+\sigma_0+\delta/2+o(1)} \quad \textit{and} \quad J \ge X^{19/68+\sigma_0+\delta/2+o(1)}. \tag{46}$$

The second value inherits the same version lock as the Li-v6 profile in TPC-20.

Proof. From $QJ \asymp X$,

$$Q^2X^{\varepsilon} \asymp \frac{XQ}{J} X^{\varepsilon}.$$

Condition (43) gives (45). Choose ε smaller than a fixed fraction of $1/2-\delta+\eta_0$. The resulting fixed power saving dominates every prescribed power of $\log X$, proving (44).

For the profile ledger, write $L = X^{\lambda+o(1)}$ and $D \le X^{1/4-\delta/2+o(1)}$. Then

$$J \asymp X/(LD) \ge X^{3/4-\lambda+\delta/2+o(1)}.$$

Substitution of $\lambda = 10/21-\sigma_0$ and $\lambda = 8/17-\sigma_0$ gives (46). ☐

Remark 3.5 (What has been closed)*.* The estimate is absolute in the lcm denominator and uses no cancellation between distinct q's. It closes the explicit Ramanujan mean of the chosen compatible short–short packet. It does not estimate the centered fiber discrepancy, restore either large-divisor leg, or bound the full primitive residual correlation.

4 Deterministic row occupancy and the long-lcm frontier

This section uses one feature which is absent from an arbitrary fiber vector: the fibers come from rows $m = \ell d$, with $\ell \asymp L$, $d \asymp D$, and $L > 2R$. It gives a genuine anti-concentration estimate, but the resulting discrepancy bound reaches only the already-completable side of the joint-period boundary. The Ramanujan mean is handled separately and more strongly in section 3.

4.1 Safe restoration of the Cartesian row packet

Put

$$Z_m^{\mathrm{sh}}(j) = \sum_{\substack{u \le R \\ (u,mH)=1}} b_R(u) \left(\mathbf{1}_{u|mj+h} - \frac{1}{u} \right). \tag{47}$$

Only the short coefficient $b_R(u)\mathbf{1}_{u\le R}$ is used here.

Lemma 4.1 (Pointwise size of the isolated short signal)**.** *Uniformly on the TPC row and orbit support, for every fixed $\epsilon > 0$,*

$$|Z_m^{\mathrm{sh}}(j)| \ll_{\epsilon,h} X^{\epsilon}. \tag{48}$$

Proof. The coefficient is squarefree supported and

$$\max_{u \le R} |b_R(u)| \ll (\log(2R))^2, \qquad \sum_{u \le R} \frac{|b_R(u)|}{u} \ll (\log(2R))^2. \tag{49}$$

Indeed, the first estimate follows from the positive formula for λ_R', while the second follows from $b_R(u) = -\mu(u)\log u - \lambda_R'(u)$ and the Euler-product estimate $\sum_{u\le R} |\lambda_R'(u)|/u \ll (\log(2R))^2$. Consequently

$$|Z_m^{\mathrm{sh}}(j)| \le \tau(mj+h) \max_{u\le R} |b_R(u)| + \sum_{u \le R} \frac{|b_R(u)|}{u}.$$

The targets are $O(X)$, so the standard divisor bound proves (48). $\square$

Proposition 4.2 (Scalar-safe unpruning)**.** *Fix $\kappa, \lambda_0 > 0$, assume $L \ge X^{\lambda_0}$, and let the row weights be polylogarithmically bounded. In a scalar short–short correlation, the diagonal, same-source, near-determinant, and large row-gcd pairs may be restored at total cost*

$$O_\epsilon\Big(X^{1+\epsilon} + XQ\{L^{-1} + X^{-\kappa}\}X^{\epsilon} \Big). \tag{50}$$

In particular, after taking $\epsilon > 0$ sufficiently small, this is $O_A(XQ(\log X)^{-A})$ for every fixed $A > 0$.

Proof. Apply lemma 4.1 to both short signals and repeat the tuple counts in the TPC-18 degenerate-channel removal [5]. The diagonal has $O(LDJ) = O(X)$ tuples. The same-source channel has $O(LD^2J) = O(XQ/L)$ tuples, the near channel has $O(XQ(X^{-\kappa} + L^{-1}))$, and the large-gcd channel has $O(XQX^{-\kappa})$. Source logarithms and bounded smooth weights are absorbed into X^{ϵ}. Since $Q \ge L \ge X^{\lambda_0}$, the diagonal also has a fixed power saving relative to XQ. $\square$

Remark 4.3 (The order of operations matters). Proposition 4.2 permits restoration of the full Cartesian row set in the *total scalar short–short packet*. It does not say that every individual q-fiber, Poisson frequency, or mean/discrepancy piece of the excluded rows is small. Thus one must unprune before the fiber decomposition and keep (50) outside it; coefficientwise unpruning would be an unjustified stronger assertion.

4.2 A row-fiber occupancy theorem

Let $\mathscr{A}$ be the full row family

$$\alpha = (\ell, d), \qquad \ell \in [L/2, 2L] \text{ prime}, \qquad d \in [D, 2D] \cap [1, V], \qquad m_\alpha = \ell d, \tag{51}$$

with any rowwise coprimality masks inserted into the coefficients. A smooth separation of the original two-row weight gives bilinear packets $x_\alpha y_\beta F(j/J)$. We assume

$$|x_\alpha| \le W_x, \qquad |y_\alpha| \le W_y, \tag{52}$$

where W_x, W_y are polylogarithmic in the application. For squarefree $u \le R$, $(u, H) = 1$, define

$$A_u^x(\kappa) = \sum_{\substack{\alpha \in \mathscr{A} \\ m_\alpha \kappa \equiv -h \pmod{u}}} x_\alpha, \qquad \kappa \in \mathcal{U}_u, \tag{53}$$

and define A_u^y similarly.

Lemma 4.4 (Deterministic row anti-concentration)**.** *Uniformly for $u \le R$,*

$$\|A_u^x\|_\infty \ll W_x \frac{Q}{u}, \tag{54}$$

$$\sum_{\kappa \in \mathcal{U}_u} |A_u^x(\kappa)| \ll W_x Q, \tag{55}$$

$$\|A_u^x\|_2 \ll W_x \frac{Q}{\sqrt{u}}. \tag{56}$$

The same estimates hold with x replaced by y.

Proof. Fix d and κ. The congruence in (53) confines ℓ to one class modulo u. An interval of length $O(L)$ contains $O(L/u + 1) = O(L/u)$ integers in that class because $L > 2R \ge 2u$. There are $O(D)$ choices of d, proving (54). Every admissible row belongs to exactly one class κ, so summing absolute values before grouping proves (55). Finally, $\|A\|_2^2 \le \|A\|_\infty \|A\|_1$, which gives (56). □

Let ϑ be supported on squarefree $u \le R$, and suppose

$$|\vartheta(u)| \le \Theta_R. \tag{57}$$

Both $\lambda'_R(u)$ and $b_R(u)\mathbf{1}_{u \le R}$ satisfy this with $\Theta_R \ll (\log(2R))^2$. For squarefree q with $(q, H) = 1$, define the full bilinear lcm fiber

$$Z_q^{x,y}(\kappa) = \sum_{\substack{u,v \le R \\ [u,v] = q}} \vartheta(u)\vartheta(v) A_u^x(\kappa \bmod u) A_v^y(\kappa \bmod v), \qquad \kappa \in \mathcal{U}_q. \tag{58}$$

No complex conjugation is implicit in (58); after Mellin separation the packet is bilinear. A Hermitian square is the special case in which the second factor is conjugated.

Theorem 4.5 (Actual-fiber sup norm and variance)**.** *Put*

$$\mathfrak{g}(q)=\prod_{p\mid q}\left(2+\frac{1}{p}\right). \tag{59}$$

Then

$$\|Z_q^{x,y}\|_\infty \ll W_xW_y\Theta_R^2Q^2\frac{\mathfrak{g}(q)}{q}, \tag{60}$$

$$|\overline{Z}_q^{x,y}| \ll W_xW_y\Theta_R^2Q^2\frac{\mathfrak{g}(q)}{q}, \tag{61}$$

$$\|D_q^{x,y}\|_2 \ll W_xW_y\Theta_R^2Q^2\frac{\mathfrak{g}(q)}{\sqrt{q}}, \tag{62}$$

where $D_q^{x,y}=Z_q^{x,y}-\overline{Z}_q^{x,y}$ and the mean is over $\mathcal{U}_q$. The same three estimates hold after multiplying each row pair in (58) by any Schur mask of modulus at most 1, in particular by the generic mask (17).

Proof. For squarefree q, independent assignment of every prime divisor to u only, v only, or both gives the exact identity

$$\sum_{\substack{u,v\mid q\\ [u,v]=q}}\frac{1}{uv}=\prod_{p\mid q}\left(\frac{2}{p}+\frac{1}{p^2}\right)=\frac{1}{q}\prod_{p\mid q}\left(2+\frac{1}{p}\right). \tag{63}$$

The restrictions $u,v\le R$ can only decrease the positive sum. Insert (54) in (58) and use (63) to obtain (60). If a Schur mask $\mathfrak{m}(\alpha,\beta)$ with $|\mathfrak{m}|\le 1$ is inserted, the absolute value of its (u,v,κ)-fiber is at most

$$\left(\sum_{\alpha\text{ in the }u\text{-fiber}}|x_\alpha|\right)\left(\sum_{\beta\text{ in the }v\text{-fiber}}|y_\beta|\right),$$

so exactly the same majorant applies. The mean bound is immediate. Finally, subtraction of the mean is orthogonal in $\ell^2(\mathcal{U}_q)$, and therefore

$$\|D_q^{x,y}\|_2\le\|Z_q^{x,y}\|_2\le\sqrt{\varphi(q)}\,\|Z_q^{x,y}\|_\infty.$$

This proves (62). □

4.3 What the occupancy bound buys

Let $q\asymp Y$ be a dyadic denominator block and retain the low-frequency window of TPC-20. Assume $J\ge 4HX^{\epsilon_0}$, so that the positive and negative retained frequencies do not wrap modulo q, and assume the polynomial fiber-mass condition needed to remove the rapidly decaying Fourier tail.

Corollary 4.6 (Dyadic discrepancy consequence)**.** *For fixed H, with a summed Fourier tail smaller than every power of X,*

$$\sum_{q\asymp Y}|\mathcal{L}_q^{\mathrm{disc}}|\ll_{F,H,\epsilon_0}X^{\epsilon_0/2}W_xW_y\Theta_R^2Q^2\sqrt{JY}\log(2Y), \tag{64}$$

Relative to the natural scale $XQ=Q^2J$, the right-hand side is

$$\ll X^{\epsilon_0/2}W_xW_y\Theta_R^2\log(2Y)\sqrt{\frac{Y}{J}}. \tag{65}$$

Consequently this method gives a fixed power saving when $Y\le JX^{-\eta}$, after taking $0<\epsilon_0<\eta$, but gives no fixed power saving in a genuinely long block $Y\ge JX^{\eta}$.

Proof. The restricted-frequency Parseval estimate of TPC-20 gives

$$|\mathcal{L}_q^{\mathrm{disc}}| \ll X^{\epsilon_0/2}\sqrt{J}\,\|D_q^{x,y}\|_2.$$

Moreover, for $\Re s > 1$,

$$\sum_{q\geq 1} \frac{\mu^2(q)\mathfrak{g}(q)}{q^s} = \zeta(s)^2\mathcal{H}(s),$$

where the local factor of $\mathcal{H}$ is

$$\left(1+\left(2+\frac{1}{p}\right)p^{-s}\right)(1-p^{-s})^2.$$

After the linear terms $2p^{-s}$ cancel, the remaining Euler product is absolutely convergent for $\Re s > 1/2$. If $\mathcal{H}(s) = \sum_n h(n)n^{-s}$, then in particular $\sum_n |h(n)|/n < \infty$. Dirichlet convolution with the divisor function, followed by $\sum_{n\leq T}\tau(n) \ll T\log(2T)$, gives

$$\sum_{q\leq T}\mu^2(q)\mathfrak{g}(q) \ll T\log(2T), \qquad \sum_{q\asymp Y}\frac{\mu^2(q)\mathfrak{g}(q)}{\sqrt{q}} \ll \sqrt{Y}\log(2Y). \tag{66}$$

Sum (62) and use (66). Division by Q^2J proves (65). □

Remark 4.7 (A normalization gain, not a long-lcm estimate). The factor $q^{-1/2}$ in (62) is a genuine improvement over treating $Z_q(\kappa)$ as an arbitrary fiber vector. Nevertheless, (65) crosses the natural scale precisely at $Y \asymp J$, which is also the joint-period completion boundary up to fixed powers of X and the fixed factor H. Thus deterministic row occupancy does not penetrate the long-lcm gate. This is a limit of the stated estimate, not a lower bound for the arithmetic packet.

The mean term behaves better: theorem 3.3 closes it absolutely for the original masked packet, without restoring the Cartesian row family or invoking the occupancy estimate.

5 Divisor-lattice reassembly of rank-one fibers

This section records the exact extra structure present before the generic row-pair mask is imposed. It applies to one separated rank-one packet; the two row weights need not be complex conjugates. The distinction is important because smooth separation preserves bilinear rank one, but it does not make every individual separated packet positive semidefinite. All divisor layers in this section are admissible and hence coprime to H; identically zero inadmissible layers are omitted.

For $u \leq R$ and $\xi \in \mathcal{U}_u$, define

$$A_u^x(\xi) = \sum_{\substack{\alpha\in\mathcal{A}\\ m_\alpha\equiv -h\bar{\xi}\ (u)}} x_\alpha, \qquad A_u^y(\xi) = \sum_{\substack{\alpha\in\mathcal{A}\\ m_\alpha\equiv -h\bar{\xi}\ (u)}} y_\alpha. \tag{67}$$

The congruence forces $(m_\alpha, u) = 1$. If $u \mid s$, we lift a function on $\mathcal{U}_u$ to $\mathcal{U}_s$ through reduction modulo u, without changing its notation. Put

$$B_s^x(\xi) = \sum_{u|s}\vartheta_R(u)A_u^x(\xi \bmod u), \qquad B_s^y(\xi) = \sum_{u|s}\vartheta_R(u)A_u^y(\xi \bmod u). \tag{68}$$

The coefficient ϑ_R is extended by zero, so divisors outside the chosen short support make no contribution.

Theorem 5.1 (Lcm layers are divisor Möbius differences)**.** *For the unmasked packet* $\mathfrak{m} \equiv 1$, *one has pointwise on* $\mathcal{U}_q$

$$\boxed{Z_q^1(\kappa) = \sum_{s|q} \mu(q/s) B_s^x(\kappa \bmod s) B_s^y(\kappa \bmod s).} \tag{69}$$

Equivalently,

$$\sum_{q|s} Z_q^1(\xi \bmod q) = B_s^x(\xi) B_s^y(\xi). \tag{70}$$

In the Hermitian case $y_\alpha = \overline{x_\alpha}$, *the right side of* (70) *is* $|B_s^x(\xi)|^2 \geq 0$.

Proof. For fixed u, v, the two row sums separate:

$$Z_{u,v}^1(\kappa) = A_u^x(\kappa \bmod u) A_v^y(\kappa \bmod v).$$

Therefore

$$\begin{aligned} B_s^x(\xi) B_s^y(\xi) &= \sum_{u,v|s} \vartheta_R(u) \vartheta_R(v) A_u^x(\xi) A_v^y(\xi) \\ &= \sum_{q|s} Z_q^1(\xi \bmod q), \end{aligned}$$

because $u, v \mid s$ is equivalent to $[u, v] \mid s$. Ordinary Möbius inversion on the divisor lattice proves (69). $\square$

5.1 Means, centered prefix products, and covariance

Define

$$E_s^{x,y} = \frac{1}{\varphi(s)} \sum_{\xi \in \mathcal{U}_s} B_s^x(\xi) B_s^y(\xi), \qquad d_s^{x,y}(\xi) = B_s^x(\xi) B_s^y(\xi) - E_s^{x,y}. \tag{71}$$

The reduction map $\mathcal{U}_q \to \mathcal{U}_s$ has exactly $\varphi(q)/\varphi(s)$ elements in every fiber when $s \mid q$. Taking means and then centering (69) gives the following exact consequence.

Corollary 5.2 (Divisor lift of the unmasked discrepancy)**.** *For every squarefree* q,

$$\overline{Z}_q^1 = \sum_{s|q} \mu(q/s) E_s^{x,y}, \tag{72}$$

$$D_q^1(\kappa) = \sum_{s|q} \mu(q/s) d_s^{x,y}(\kappa \bmod s). \tag{73}$$

In the Hermitian case $E_s^{x,\overline{x}} \geq 0$, *and*

$$\sum_{q|s} \overline{Z}_q^1 = E_s^{x,\overline{x}} \geq 0. \tag{74}$$

No positivity is asserted for an individual lcm layer.

For $s_1, s_2 \mid q$, put $w = [s_1, s_2]$ and define the normalized prefix covariance

$$\Gamma_{s_1,s_2} = \frac{1}{\varphi(w)} \sum_{\xi \in \mathcal{U}_w} d_{s_1}^{x,y}(\xi \bmod s_1) \overline{d_{s_2}^{x,y}(\xi \bmod s_2)}. \tag{75}$$

Expanding the square in (73) gives the exact variance formula

$$\boxed{\|D_q^1\|_{2,\mathcal{U}_q}^2 = \varphi(q) \sum_{s_1,s_2|q} \mu(q/s_1) \mu(q/s_2) \Gamma_{s_1,s_2}.} \tag{76}$$

Thus even the unmasked variance is a connected divisor covariance, not the sum of independent prefix variances.

5.2 Additive Fourier lifting and primitive sign loss

Use the additive transform

$$\widehat{d}_s^{x,y}(a) = \sum_{\xi \in \mathcal{U}_s} d_s^{x,y}(\xi) e_s(a\overline{H}\xi). \tag{77}$$

When $s = 1$, the centered function is zero. For $s \mid q$, put $t = q/s$; because q is squarefree, $(s,t) = 1$. Let $\bar{t}$ denote the inverse of $t \pmod{s}$.

Theorem 5.3 (CRT lift of the divisor discrepancy)**.** *For every $r \in \mathbb{Z}$,*

$$\boxed{\widehat{D}_q^1(r) = \sum_{s|q} \mu(t) c_t(r) \widehat{d}_s^{x,y}(r\bar{t} \bmod s), \qquad t = q/s.} \tag{78}$$

Since t is squarefree,

$$\mu(t)c_t(r) = \mu((t,r))\varphi((t,r)). \tag{79}$$

In particular, if $(r,q) = 1$, then

$$\boxed{\widehat{D}_q^1(r) = \sum_{s|q} \widehat{d}_s^{x,y}(r\overline{q/s} \bmod s).} \tag{80}$$

All outer lcm Möbius signs have disappeared. The summands may still cancel through their internal phases; no positivity of the Fourier coefficients is claimed.

Proof. Lift one function $d_s^{x,y}$ from $\mathcal{U}_s$ to $\mathcal{U}_q$. CRT factorization with $q = st$ gives

$$\begin{aligned}
&\sum_{\kappa \in \mathcal{U}_q} d_s^{x,y}(\kappa \bmod s) e_q(r\overline{H}\kappa) \\
&\quad = \widehat{d}_s^{x,y}(r\bar{t} \bmod s) \sum_{\eta \in \mathcal{U}_t} e_t(r\overline{H}\bar{s}\eta) \\
&\quad = c_t(r)\widehat{d}_s^{x,y}(r\bar{t} \bmod s).
\end{aligned}$$

Insert (73). For squarefree t, the standard formula $c_t(r) = \mu(t/(t,r))\varphi((t,r))$ proves (79) and the primitive-frequency specialization. □

The same lifting gives a useful but generally lossy sufficient bound:

$$\boxed{\|D_q^1\|_{2,\mathcal{U}_q} \le \sum_{s|q} \sqrt{\frac{\varphi(q)}{\varphi(s)}} \|d_s^{x,y}\|_{2,\mathcal{U}_s}.} \tag{81}$$

It follows by applying the triangle inequality to (73); the norm of a lifted function is multiplied by $\sqrt{\varphi(q)/\varphi(s)}$.

5.3 Reduction to one-row second and fourth moments

Let

$$C_s^x = \frac{1}{\varphi(s)} \sum_{\xi \in \mathcal{U}_s} B_s^x(\xi), \qquad \mathcal{E}_s^x(\xi) = B_s^x(\xi) - C_s^x, \tag{82}$$

and define $C_s^y, \mathcal{E}_s^y$ analogously. Direct expansion gives

$$d_s^{x,y} = C_s^x \mathcal{E}_s^y + C_s^y \mathcal{E}_s^x + \mathcal{E}_s^x \mathcal{E}_s^y - \frac{1}{\varphi(s)} \sum_{\xi \in \mathcal{U}_s} \mathcal{E}_s^x(\xi)\mathcal{E}_s^y(\xi). \tag{83}$$

Proposition 5.4 (One-row moment transfer)**.** *With unnormalized $L^p(\mathcal{U}_s)$ norms,*

$$\boxed{\|d_s^{x,y}\|_2 \le |C_s^x|\|\mathcal{E}_s^y\|_2 + |C_s^y|\|\mathcal{E}_s^x\|_2 + 2\|\mathcal{E}_s^x\|_4\|\mathcal{E}_s^y\|_4.} \tag{84}$$

In the Hermitian case this becomes

$$\||B_s^x|^2 - E_s^{x,\overline{x}}\|_2 \le 2|C_s^x|\|\mathcal{E}_s^x\|_2 + 2\|\mathcal{E}_s^x\|_4^2. \tag{85}$$

Proof. Only the centered product in (83) needs attention. Hölder gives $\|\mathcal{E}_s^x\mathcal{E}_s^y\|_2 \le \|\mathcal{E}_s^x\|_4\|\mathcal{E}_s^y\|_4$. Moreover,

$$\sqrt{\varphi(s)}\left|\frac{1}{\varphi(s)}\sum_\xi \mathcal{E}_s^x(\xi)\mathcal{E}_s^y(\xi)\right| \le \|\mathcal{E}_s^x\|_4\|\mathcal{E}_s^y\|_4.$$

The triangle inequality proves the claim. □

The row means and errors in this proposition have a direct arithmetic interpretation. Averaging (67) over $\xi \in \mathcal{U}_u$ yields

$$\frac{1}{\varphi(u)}\sum_{\xi\in\mathcal{U}_u} A_u^x(\xi) = \frac{1}{\varphi(u)}\sum_{\substack{\alpha\in\mathcal{A}\\(m_\alpha,u)=1}} x_\alpha. \tag{86}$$

For the TPC rows, fixing d turns $A_u^x(\xi) - \operatorname{avg} A_u^x$ into a prime-progression error for

$$\ell \equiv -h\overline{\xi}\,\overline{d} \pmod{u}. \tag{87}$$

Thus (84) is a precise point of entry for progression second and fourth moments. It is not itself such an estimate.

Remark 5.5 (Mask boundary)*.* The divisor-lattice formulas in this section require a rank-one unmasked row packet. A Schur mask depending jointly on (m_1, m_2) does not factor through (67). The generic mask is treated without deletion in section 6. Scalar degenerate removal does not justify inserting $\mathfrak{m} \equiv 1$ into the nonlinear norm (25); section 7 gives the exact counterexample.

6 The shared-factor character gate

The additive transform in (22) is naturally defined on the full cyclic group modulo q. The coefficient itself, however, is supported on the unit group. Multiplicative Fourier analysis on that unit group retains the provenance of both rows and, in particular, keeps the generic Schur mask inside the transform. This section gives the exact change of basis. It is an identity, not an estimate for the resulting character moment.

For a positive integer s, write

$$G_s = (\mathbb{Z}/s\mathbb{Z})^\times, \qquad \mathfrak{X}(s) = \widehat{G}_s,$$

with G_1 and $\mathfrak{X}(1)$ interpreted as trivial groups. A character is extended by zero away from the units whenever it is evaluated on an integer. For $f : G_s \to \mathbb{C}$, use the convention

$$\widehat{f}^\times(\chi) = \sum_{k\in G_s} f(k)\overline{\chi(k)}. \tag{88}$$

Then

$$f(k) = \frac{1}{\varphi(s)}\sum_{\chi\in\mathfrak{X}(s)} \widehat{f}^\times(\chi)\chi(k), \tag{89}$$

$$\sum_{k\in G_s}|f(k)|^2=\frac{1}{\varphi(s)}\sum_{\chi\in\mathfrak{X}(s)}|\widehat{f}^{\times}(\chi)|^2. \tag{90}$$

In particular, if $f^\circ=f-|G_s|^{-1}\sum_{k\in G_s}f(k)$, then

$$\boxed{\|f^\circ\|_{2,G_s}^2=\frac{1}{\varphi(s)}\sum_{\substack{\chi\in\mathfrak{X}(s)\\ \chi\neq\mathbf{1}_s}}|\widehat{f}^{\times}(\chi)|^2.} \tag{91}$$

Thus "connected" below means that the principal unit-group character, equivalently the fiber mean, has been removed.

6.1 The masked shared-g convolution

Fix squarefree divisor moduli $u,v\le R$ with $(uv,H)=1$. Modulus pairs failing this coprimality condition have empty admissible fibers and are omitted. Compatibility of the two row labels is imposed inside the fiber. Write

$$g=(u,v),\qquad u=ga,\qquad v=gb,\qquad q=[u,v]=gab. \tag{92}$$

The integers g,a,b are pairwise coprime. CRT therefore identifies $G_q=G_g\times G_a\times G_b$. Write a character of G_q as

$$\chi=\chi_g\boxtimes\chi_a\boxtimes\chi_b,\qquad \chi_c\in\mathfrak{X}(c).$$

For characters $\alpha\in\mathfrak{X}(u)$ and $\beta\in\mathfrak{X}(v)$, define the masked two-row Dirichlet-character sum

$$\mathcal{B}^{u,v}_{\mathfrak{m}}(\alpha,\beta)=\sum_{\omega_1,\omega_2\in\mathcal{A}}x_{\omega_1}y_{\omega_2}\mathfrak{m}(\omega_1,\omega_2)\alpha(m_{\omega_1})\beta(m_{\omega_2}). \tag{93}$$

Here $\omega_i=(\ell_i,d_i)$ is a row label. The zero extension of the characters inserts the row–modulus coprimality conditions. There is no compatibility indicator in (93); it will be created exactly by averaging a character of the shared factor g.

Theorem 6.1 (Exact masked shared-factor transform)**.** *With the conventions above,*

$$\boxed{\widehat{Z}^{\mathfrak{m},\times}_{u,v}(\chi)=\frac{\overline{\chi(-h)}}{\varphi(g)}\sum_{\psi\in\mathfrak{X}(g)}\mathcal{B}^{u,v}_{\mathfrak{m}}\left((\chi_g\psi)\boxtimes\chi_a,\overline{\psi}\boxtimes\chi_b\right).} \tag{94}$$

The factor $1/\varphi(g)$ is essential. It is the normalized orthogonality projector onto compatible row pairs.

Proof. Suppose first that a pair of rows contributes to $Z^{\mathfrak{m}}_{u,v}(\kappa)$. In the CRT coordinates of (92),

$$\kappa_g\equiv-h\overline{m_1}\equiv-h\overline{m_2}\pmod g,\qquad \kappa_a\equiv-h\overline{m_1}\pmod a,\qquad \kappa_b\equiv-h\overline{m_2}\pmod b.$$

Consequently

$$\overline{\chi(\kappa)}=\overline{\chi(-h)}\chi_g(m_1)\chi_a(m_1)\chi_b(m_2). \tag{95}$$

For unit rows, character orthogonality gives

$$\mathbf{1}_{\{m_1\equiv m_2\ (g)\}}=\frac{1}{\varphi(g)}\sum_{\psi\in\mathfrak{X}(g)}\psi(m_1)\overline{\psi(m_2)}. \tag{96}$$

Insert (95) and (96) in the definition of the multiplicative transform, then interchange the finite sums. The two row variables carry respectively $(\chi_g\psi)\boxtimes\chi_a$ and $\overline{\psi}\boxtimes\chi_b$, proving (94). ☐

The unmasked identity has an additional factorization. Define the one-row Dirichlet polynomials

$$\mathcal{A}_x^{(s)}(\rho) = \sum_{\omega\in\mathcal{A}} x_\omega \rho(m_\omega), \qquad \mathcal{A}_y^{(s)}(\rho) = \sum_{\omega\in\mathcal{A}} y_\omega \rho(m_\omega), \qquad \rho \in \mathfrak{X}(s). \tag{97}$$

When no ambiguity is possible, the superscript s is suppressed. The relation $k \equiv -h\overline{m} \pmod{s}$ also gives the useful one-row formula

$$\sum_{k\in G_s} A_s^x(k)\overline{\rho(k)} = \overline{\rho(-h)}\, \mathcal{A}_x^{(s)}(\rho), \tag{98}$$

and likewise for y. Indeed, $\overline{\rho(-h\overline{m})} = \overline{\rho(-h)}\rho(m)$.

Corollary 6.2 (Unmasked row-polynomial factorization)**.** *If $\mathfrak{m} \equiv 1$, then*

$$\mathcal{B}_1^{u,v}(\alpha,\beta) = \mathcal{A}_x^{(u)}(\alpha)\mathcal{A}_y^{(v)}(\beta), \tag{99}$$

$$\widehat{Z}_{u,v}^{1,\times}(\chi) = \frac{\overline{\chi(-h)}}{\varphi(g)} \sum_{\psi\in\mathfrak{X}(g)} \mathcal{A}_x^{(u)}((\chi_g\psi)\boxtimes\chi_a)\mathcal{A}_y^{(v)}(\overline{\psi}\boxtimes\chi_b). \tag{100}$$

No conjugation of the y-polynomial is implicit. In a Hermitian packet it is already contained in the choice $y_\omega = \overline{x_\omega}$.

For the generic mask (17), the factorization (99) is unavailable: the determinant, same-source, and divisor-gcd conditions jointly depend on both row labels. Formula (94) remains exact because it never removes that mask.

6.2 Lcm aggregation and the exact fourth-moment gate

Fix an admissible squarefree q, so that $(q,H)=1$. For every ordered pair $u,v \le R$ with $[u,v]=q$, use the decomposition (92) and restrict a given $\chi \in \mathfrak{X}(q)$ to the corresponding CRT factors. Define

$$\mathcal{T}_q^{\mathfrak{m}}(\chi) := \sum_{\substack{u,v\le R\\ [u,v]=q}} \frac{\vartheta_R(u)\vartheta_R(v)}{\varphi((u,v))} \sum_{\psi\in\mathfrak{X}((u,v))} \mathcal{B}_{\mathfrak{m}}^{u,v}\left((\chi_g\psi)\boxtimes\chi_a, \overline{\psi}\boxtimes\chi_b\right). \tag{101}$$

Linearity and theorem 6.1 give

$$\boxed{\widehat{Z}_q^{\mathfrak{m},\times}(\chi) = \overline{\chi(-h)}\, \mathcal{T}_q^{\mathfrak{m}}(\chi).} \tag{102}$$

Since $(q,h)=1$, the phase on the right has modulus one.

Theorem 6.3 (Exact masked connected character moment)**.** *Define*

$$\mathfrak{F}_q^{\mathfrak{m}} = \frac{1}{\varphi(q)} \sum_{\substack{\chi\in\mathfrak{X}(q)\\ \chi\neq\mathbf{1}_q}} |\mathcal{T}_q^{\mathfrak{m}}(\chi)|^2. \tag{103}$$

Then

$$\boxed{\mathfrak{F}_q^{\mathfrak{m}} = \|D_q^{\mathfrak{m}}\|_{2,\mathcal{U}_q}^2.} \tag{104}$$

Thus the inherited sufficient condition (25) is exactly a bound for

$$\sum_{q\in\mathcal{Q}} (\mathfrak{F}_q^{\mathfrak{m}})^{1/2}. \tag{105}$$

For example, with the epsilon bookkeeping of TPC-20, it suffices to prove

$$\sum_{q\in\mathcal{Q}} (\mathfrak{F}_q^{\mathfrak{m}})^{1/2} \ll X^{-\varepsilon}\frac{XQ}{\sqrt{J}}(\log X)^{-B}. \tag{106}$$

This is a sufficient input for the chosen short–short discrepancy packet, not a necessary condition for cancellation of the scalar correlation.

Proof. Apply (91) to $f = Z_q^{\mathfrak{m}}$, and then use (102). The phase $\overline{\chi(-h)}$ disappears after taking absolute values. ☐

In the unmasked case, substituting (99) into (103) makes it literally a fourth moment of one-row Dirichlet polynomials, coupled by the shared g-character convolution and the outer lcm sum. In the generic case it is the square of a masked two-row character sum, hence still a four-row moment, but not a product of two independent second moments. The word "connected" removes only the principal fiber mode. The generic row mask must remain inside $\mathcal{B}_{\mathfrak{m}}^{u,v}$; why scalar degenerate removal cannot be performed after this nonlinear norm is explained in section 7.

6.3 The Gauss–conductor bridge back to additive frequency

Put $e_s(z) = \exp(2\pi i z/s)$, and for $\chi \in \mathfrak{X}(q)$ define the generalized Gauss sum

$$G_q(\chi;t) = \sum_{k\in G_q} \chi(k)e_q(tk). \tag{107}$$

Multiplicative inversion gives the exact bridge

$$\boxed{S_q(r) = \frac{1}{\varphi(q)}\sum_{\chi\in\mathfrak{X}(q)} \overline{\chi(-h)}\mathcal{T}_q^{\mathfrak{m}}(\chi)G_q(\chi;r\overline{H}).} \tag{108}$$

For completeness, the Gauss factor in (108) can be evaluated at every integer, not merely at primitive additive frequencies.

Proposition 6.4 (Induced-character Gauss factor)**.** *Suppose that q is squarefree. Let $f \mid q$ be the conductor of χ, let χ^* (mod f) be the inducing primitive character, and put $d = q/f$. Then, for every integer t,*

$$G_q(\chi;t) = \begin{cases} \overline{\chi^*(t)}\chi^*(d)\tau_f(\chi^*)c_d(t), & (t,f)=1,\\ 0, & (t,f)>1, \end{cases} \tag{109}$$

where $\tau_f(\chi^*) = \sum_{x \bmod f}\chi^*(x)e_f(x)$, and the conductor-one conventions are $\tau_1(\mathbf{1}) = 1$ and $\chi^*(d) = 1$.

Proof. Factor the sum by CRT modulo f and d. The two factors are

$$\sum_{x\in G_f}\chi^*(x)e_f(t\overline{d}\,x) \quad\text{and}\quad \sum_{y\in G_d} e_d(t\overline{f}\,y).$$

The first vanishes when $(t,f)>1$, and otherwise equals $\overline{\chi^*(t)}\chi^*(d)\tau_f(\chi^*)$; the second is $c_d(t)$. ☐

As $(H,q)=1$, inserting (109) in (108) yields

$$S_q(r) = \frac{1}{\varphi(q)}\sum_{\substack{f\mid q\\(f,r)=1}} c_{q/f}(r)\sideset{}{^*}\sum_{\chi^*\ (\bmod f)} \chi^*(q/f)\tau_f(\chi^*)\overline{\chi^*(-hr\overline{H})}\mathcal{T}_q^{\mathfrak{m}}(\chi_q), \tag{110}$$

where χ_q is the character modulo q induced by χ^*, and the star denotes primitive characters. The $f = 1$ term is exactly

$$\frac{\mathcal{T}_q^{\mathfrak{m}}(\mathbf{1}_q)}{\varphi(q)} c_q(r) = \overline{Z}_q^{\mathfrak{m}} c_q(r), \tag{111}$$

so the terms $f > 1$ give the additive discrepancy.

The conductor normalization is important. If $(r, f) = 1$, then

$$|G_q(\chi; r\overline{H})| = \sqrt{f}\,|c_{q/f}(r)|. \tag{112}$$

In particular, for $(r, q) = 1$ the magnitude is $\sqrt{f}$, not $\sqrt{q}$ unless χ is primitive modulo q. Applying the primitive Gauss magnitude $\sqrt{q}$ to every character modulo q would therefore be incorrect.

There is also no automatic average gain hidden in the conductors. A local count at each prime $p \mid q$ gives one principal character of conductor factor 1 and $p - 2$ nonprincipal characters of conductor factor p. Hence

$$\sum_{\chi \in \mathfrak{X}(q)} \operatorname{cond}(\chi) = \prod_{p \mid q} (1 + p(p-2)) = \varphi(q)^2. \tag{113}$$

For $(t, q) = 1$, this is equivalently recovered by applying character Parseval to the unit-modulus function $k \mapsto e_q(tk)$, which gives $\sum_\chi |G_q(\chi; t)|^2 = \varphi(q)^2$. Thus a generic Cauchy–Schwarz estimate in the character variable merely transfers the full fiber energy; it does not create cancellation.

6.4 The naive large-sieve boundary

The standard multiplicative large sieve states, in one common normalization, that

$$\sum_{s \le S} \frac{s}{\varphi(s)} \sideset{}{^*}\sum_{\chi \ (\bmod\ s)} \left| \sum_{n \le N} a_n \chi(n) \right|^2 \ll (S^2 + N) \sum_{n \le N} |a_n|^2 \tag{114}$$

[3, Chapter 7]. This applies directly to one of the polynomials in (97). It does not, without further work, estimate (103): that expression contains a shared-g convolution, an lcm aggregation, the square of a bilinear row sum, and the genuinely two-row mask $\mathfrak{m}$. Dropping the mask to manufacture a product of polynomials changes the nonlinear quantity, as section 7 makes explicit.

At a fixed modulus, complete character orthogonality already gives

$$\frac{1}{\varphi(s)} \sum_{\rho \in \mathfrak{X}(s)} |\mathcal{A}_x^{(s)}(\rho)|^2 = \sum_{c \in G_s} \left| \sum_{\substack{\omega \in \mathcal{A} \\ m_\omega \equiv c\ (s)}} x_\omega \right|^2. \tag{115}$$

Thus the naive second-moment route is exactly a row-occupancy bound unless additional cancellation in the actual coefficients is proved. The unconditional occupancy estimate developed in section 4, when inserted in the inherited restricted-frequency gate, has relative size $X^{o(1)}\sqrt{Y/J}$ on $q \asymp Y$. It closes only the short-joint-period side $Y < J$ with a fixed margin and supplies no fixed-power saving in a genuinely long block $Y \ge JX^\eta$.

Consequently the character change of basis does not by itself produce a new long-lcm wedge. Such a wedge would require a theorem for the actual masked moment (103), or another arithmetic estimate that retains its row-pair mask and beats the occupancy scale by a quantified margin. No such estimate is claimed here; the large-divisor legs excluded in (14) also remain outside this reduction.

7 Degenerate scalar savings do not commute with fiber energy

The scalar unpruning statement in proposition 4.2 is linear: after exact spectral reassembly, the excluded row-pair sums may be subtracted at fixed-power cost. The sufficient gate (25) is nonlinear because it takes a fiber L^2 norm first. We now show that moving the degenerate removal through this norm is not merely unproved; it is false as a formal implication.

Assume for this section that the packet is Hermitian, $y_\alpha = \overline{x_\alpha}$, and unmasked. For a row m and squarefree q with $(q, mH) = 1$, let

$$\kappa_q(m) \equiv -h\overline{m} \pmod{q}. \tag{116}$$

Define the lcm coefficient

$$\Gamma_\vartheta(q) = \sum_{\substack{u,v \le R \\ [u,v]=q}} \vartheta_R(u)\vartheta_R(v). \tag{117}$$

The divisor-lattice identity also gives

$$\Gamma_\vartheta(q) = \sum_{s|q} \mu(q/s) \left| \sum_{\substack{u|s \\ u \le R}} \vartheta_R(u) \right|^2. \tag{118}$$

Proposition 7.1 (A row diagonal is an exact one-fiber spike)**.** *The contribution of the single row pair (m, m) to the unmasked aggregated q-fiber is*

$$\boxed{Z^{\mathrm{rowdiag}}_{q,m}(\kappa) = |x_m|^2 \Gamma_\vartheta(q) \mathbf{1}_{\{\kappa = \kappa_q(m)\}}.} \tag{119}$$

If $A = |x_m|^2 \Gamma_\vartheta(q)$, then its centered fiber satisfies

$$\|D^{\mathrm{rowdiag}}_{q,m}\|^2_{2,\mathcal{U}_q} = |A|^2 \left(1 - \frac{1}{\varphi(q)}\right), \tag{120}$$

$$\widehat{D}^{\mathrm{rowdiag}}_{q,m}(r) = A \left\{ e_q(r\overline{H}\kappa_q(m)) - \frac{c_q(r)}{\varphi(q)} \right\}. \tag{121}$$

For every nonprincipal multiplicative character χ of $\mathcal{U}_q$,

$$\left| \sum_{\kappa \in \mathcal{U}_q} D^{\mathrm{rowdiag}}_{q,m}(\kappa)\overline{\chi(\kappa)} \right| = |A|. \tag{122}$$

Thus the spike occupies every nonprincipal multiplicative mode with the same magnitude and, when $\varphi(q) > 1$, retains order-A additive discrepancy at every frequency coprime to q.

Proof. For each $u \mid q$, the row m belongs to exactly the class $\kappa_q(m)$ mod u. Hence every pair u, v with lcm q contributes at the same joint class, proving (119). Subtracting the unit-class mean $A/\varphi(q)$ and summing the two possible squared values gives (120). The additive transform is immediate. For a nonprincipal χ, the constant mean has zero transform, so the transform is $A\overline{\chi(\kappa_q(m))}$, of modulus $|A|$. □

Proposition 7.2 (The scalar estimate supplies no fiberwise saving)**.** *For every q with $\Gamma_\vartheta(q) \neq 0$ and $\varphi(q) > 1$, the single-row diagonal satisfies the exact relative identity*

$$\frac{\|D^{\mathrm{rowdiag}}_{q,m}\|_2}{|x_m|^2|\Gamma_\vartheta(q)|} = \left(1 - \frac{1}{\varphi(q)}\right)^{1/2}. \tag{123}$$

Consequently proposition 4.2, by itself, supplies no coefficientwise small factor for the row-diagonal fiber array and no bound for its sum of fiber norms. In particular, that scalar estimate does not justify restoring the full Gram fiber inside a norm, applying the triangle inequality, and subtracting the degenerate scalar error afterwards. This statement does not assert that the actual summed row-diagonal fiber norm is large; additional cancellation in $\Gamma_\vartheta(q)$ or among rows is not ruled out.

Proof. Identity (123) is (120) divided by the spike amplitude. Its right-hand side tends to 1 as $\varphi(q) \to \infty$, so the fiber norm has no uniform relative saving at the coefficient level. By contrast, the scalar estimate sums the diagonal along the physical orbit and uses the row count $LDJ \asymp X$. It contains no estimate for the coefficient array obtained after discarding the Poisson weights and taking a fiber norm. Thus the displayed triangle-inequality inference does not follow from that scalar estimate. □

The correctly pruned off-diagonal object removes the row diagonal before the norm. For one fixed u, v, put

$$C_{u,v}^{\mathrm{rowdiag}}(\kappa) = \sum_{\substack{\alpha \in \mathcal{A} \\ m_\alpha \equiv -h\bar{\kappa}\ (q)}} x_\alpha y_\alpha, \qquad q = [u, v], \tag{124}$$

with the summand restricted to rows admissible for both divisor legs. Then $Z_{u,v}^{\neq} = Z_{u,v}^{1} - C_{u,v}^{\mathrm{rowdiag}}$ contains only distinct rows. Its centered norm is nonnegative and has the multiplicative Parseval formula of section 6. Subtracting a scalar self-energy from an already squared raw norm would not give the same quantity and need not remain nonnegative.

Proposition 7.3 (Distinctness alone does not force fiber spread)**.** *Even after row-diagonal subtraction, rank-one provenance and off-diagonal support alone imply no uniform fiber saving. Fix one admissible divisor pair u, v, put $g = (u, v)$, $q = [u, v]$, and take one distinct compatible row pair $m \neq n$ with $g \mid m - n$. If*

$$A = \vartheta_R(u)\vartheta_R(v)x_m y_n$$

is its isolated $(u, v; m, n)$ amplitude, then the contribution is A at exactly one joint class and zero elsewhere. After centering, its fiber L^2 energy is exactly

$$|A|^2 \left(1 - \frac{1}{\varphi(q)}\right). \tag{125}$$

Consequently a saving must use the distribution and signs of the full arithmetic row family, not only the labels Gram *or* off diagonal.

Remark 7.4 (The legitimate order)*.* There are two legitimate routes. One may retain the generic Schur mask inside the exact character transform and estimate its connected moment. Alternatively, one may reassemble the complete scalar spectrum first, estimate its rank-one part without taking a lossy raw fiber norm, and then subtract the scalar degenerate contributions using proposition 4.2. What is forbidden is to replace the generic fiber by the unpruned fiber inside a nonlinear norm merely because the excluded physical correlation is small.

8 The quantitative character-moment gate

The preceding sections remove the deterministic mean and identify two coefficient-independent bounds for the discrepancy. This section records the exact improvement still required in a long dyadic lcm block and compares it with standard character and Kloosterman technology.

8.1 The gain ledger

Write

$$q \asymp Y = X^{y+o(1)}, \qquad J = X^{j+o(1)}, \qquad Q = X^{\beta+o(1)}, \qquad j = 1-\beta. \tag{126}$$

The occupancy estimate gives, up to logarithms,

$$\sum_{q \asymp Y} \|D_q^{\mathrm{m}}\|_{2,\mathcal{U}_q} \ll Q^2 \sqrt{Y}\, X^{\varepsilon}. \tag{127}$$

Suppose a future arithmetic estimate improves this by a fixed power:

$$\sum_{q \asymp Y} \|D_q^{\mathrm{m}}\|_{2,\mathcal{U}_q} \ll Q^2 \sqrt{Y}\, X^{-\omega+\varepsilon}. \tag{128}$$

Proposition 8.1 (Gain threshold for the sufficient norm route)**.** *The combination of* (128) *with the TPC-20 restricted-frequency inequality gives a fixed power saving relative to* $XQ = Q^2 J$ *whenever*

$$\boxed{\omega > \frac{y-j}{2}} \tag{129}$$

with a fixed margin. At the maximal short–short denominator

$$Y = R^2 = X^{1-2\delta+o(1)}, \tag{130}$$

condition (129) *becomes*

$$\omega > \frac{\beta - 2\delta}{2}. \tag{131}$$

On the largest divisor slice, $\beta = \lambda + 1/4 - \delta/2$*, this is*

$$\boxed{\omega > \frac{\lambda}{2} + \frac{1}{8} - \frac{5\delta}{4}.} \tag{132}$$

These are requirements for this sufficient norm route, not lower bounds on every possible scalar method.

Proof. Equations (25) and (128) give

$$\sum_{q \asymp Y} |\mathcal{L}_q^{\mathrm{disc}}| \ll Q^2 \sqrt{JY}\, X^{-\omega+\varepsilon}.$$

After division by $Q^2 J$, the remaining factor is $X^{(y-j)/2-\omega+\varepsilon}$. This is power-saving precisely under (129) with a fixed margin. Substitute $y = 1-2\delta$, $j = 1-\beta$, and then the displayed largest-slice value of β. □

For the published TPC profile $\lambda = 10/21 - \sigma_0$, the right side of (132) is

$$\frac{61}{168} - \frac{\sigma_0}{2} - \frac{5\delta}{4}.$$

For the version-locked Li-v6 profile $\lambda = 8/17 - \sigma_0$, it is

$$\frac{49}{136} - \frac{\sigma_0}{2} - \frac{5\delta}{4}.$$

The symbol σ_0 here is the inherited profile margin, while ω denotes the new character-moment gain. Neither displayed gain is proved in this paper.

8.2 What the ordinary character large sieve returns

The classical multiplicative large sieve in primitive-character form is recorded in (114); see also Iwaniec and Kowalski [3, Chapter 7]. Passing between all and primitive characters costs only the usual divisor bookkeeping in the present squarefree setting.

For a one-row TPC polynomial, the ambient length is $N \asymp Q$ and the coefficient energy is $O(QX^{\varepsilon})$. When $U \le \sqrt{Q}$, (114) therefore has size Q^2X^{ε}, the same normalization already supplied by deterministic occupancy. When $U > \sqrt{Q}$, its U^2 term is larger. Applying Cauchy–Schwarz to the shared-g convolution in theorem 6.1 introduces a fourth moment rather than removing this cost. Thus the ordinary large sieve, used without additional arithmetic structure, proves no $\omega > 0$ in a block with $Y > J$.

This comparison does not say that character methods are irrelevant. It says that the required input must exploit the specific masked convolution over the common factor g, rather than apply the one-polynomial large sieve independently to every factor.

8.3 Why one-fraction black boxes do not yet apply

The bilinear Kloosterman-fraction estimate of Duke et al. [2] and the trilinear estimate of Bettin and Chandee [1] take separated coefficient sequences. The partially fixed-modulus preprint of Wright [8] retains an analogous separability requirement. The generic transform in theorem 6.1 contains instead

$$\mathcal{B}_{\mathfrak{m}}(\rho_1, \rho_2) = \sum_{\alpha_1, \alpha_2} x_{\alpha_1} y_{\alpha_2} \mathfrak{m}(\alpha_1, \alpha_2) \rho_1(m_1) \rho_2(m_2),$$

where the determinant and gcd restrictions remain inside the two-row coefficient. Replacing this by a product of one-row polynomials would delete the generic Schur mask before a nonlinear moment is taken, an operation not justified by the scalar estimate; see proposition 7.2.

Likewise, the exceptional-spectrum large sieve of Pascadi [4] requires a representing measure satisfying its size and two-frequency concentration hypotheses. The exact character normal form derived here does not by itself verify those hypotheses. None of these cited results is used as a black-box estimate for (128).

8.4 Three defensible next routes

The remaining short–short problem now has three explicit routes.

(i) Prove (128) directly for the masked shared-g transform, with the gain ledger (129).

(ii) Decompose the generic Schur mask into separated pieces with a fully controlled projective norm, then insert the exact parameters into one specified character or Kloosterman theorem. Merely counting pieces is insufficient; the complete norm cost must retain a fixed margin after (129).

(iii) Avoid the sufficient fiber norm and estimate the complete scalar spectral sum while preserving cancellation between the connected piece and the explicitly removable degenerate pieces. This route must keep the scalar order of operations in remark 7.4.

Even a successful short–short estimate leaves the residual legs with $u > R$ or $v > R$. Closing the full TPC-18 residual would additionally require uniform reassembly of the separated packets and all earlier endpoint estimates. Nothing in this ledger is a twin-prime lower bound, a Hardy–Littlewood asymptotic, or a breach of the sieve parity barrier.

9 Finite spectral meaning, certificate, and conclusion

9.1 Two finite dualities

The word *spectrum* has two finite meanings in this paper. After zero extension from $\mathcal{U}_q$ to $\mathbb{Z}/q\mathbb{Z}$, the additive transform

$$r \longmapsto \sum_{\kappa\in\mathcal{U}_q} D_q(\kappa)e_q(r\overline{H}\kappa)$$

is the spectral coordinate of the observable D_q for translation on the finite cycle $\mathbb{Z}/q\mathbb{Z}$. The multiplicative transform

$$\chi \longmapsto \sum_{\kappa\in\mathcal{U}_q} D_q(\kappa)\overline{\chi(\kappa)}$$

is Fourier analysis on the finite abelian group $\mathcal{U}_q$. The former couples directly to Poisson summation; the latter exposes the row Dirichlet polynomials and the shared-g convolution. The induced character Gauss formula in proposition 6.4 is the exact bridge between them.

These are literal spectra of finite group actions. They do not assert that zeros of the Riemann zeta function are eigenvalues of a dynamical system, and no such interpretation is used anywhere in the proofs.

9.2 Exact computational certificate

The standard-library certificate is

`experiments/tpc21_certificate.py`.

Run it from the paper directory with

```
python experiments/tpc21_certificate.py
python -O experiments/tpc21_certificate.py
```

Both commands regenerate `experiments/tpc21_certificate.json`. The certificate performs 1,149 exact equality checks. Pass/fail checks use integers, rational arithmetic, and exact roots of unity in a finite prime field; floating-point comparisons are not used for the proof identities. The certificate verifies:

(i) compatible row-fiber totals and the unit-class mean;

(ii) the squarefree lcm occupancy product;

(iii) bilinear divisor-lattice Möbius reassembly, centering, and the additive CRT lift;

(iv) primitive-frequency loss of the outer Möbius signs;

(v) multiplicative-character Parseval and the exact shared-g masked character convolution;

(vi) the conductor-sensitive induced-character Gauss formula;

(vii) the row-diagonal one-fiber spike and its centered energy; and

(viii) the rational exponent-gain ledger.

The generated JSON explicitly records that these finite checks are not evidence for a twin-prime theorem, the residual-dispersion estimate, a parity breakthrough, or an asymptotic prime-pair formula. For the archived version, the JSON SHA-256 is `21AB51D5A8775832D8C44D90B4E21917C11FADAD1F150B896D234835740D5F74`, and the script SHA-256 is `26268E848E0D66A3B50B6FD628E95D8AF6931DE3FA9FA31D55C722FB8BFB3CB3`. The same hashes are recorded in the README.

9.3 Conclusion

TPC-20 left a Ramanujan mean packet and a centered CRT-fiber discrepancy. The first is now closed. Reduced-class Poisson summation identifies it with an ordinary smooth coprime-lattice discrepancy of size $O_F(\tau(Hq))$, and the full lcm arithmetic cost is summable. The total mean is $O_\varepsilon(Q^2X^\varepsilon)$, one complete orbit-length factor below the natural scale.

For the discrepancy, deterministic row occupancy gives the genuine normalization $Q^2q^{-1/2}$, but its dyadic sum crosses the target exactly at $q \asymp J$. It therefore does not penetrate the long-lcm range. The new exact normal forms explain what extra information is needed. Unmasked rank-one packets are divisor-lattice Möbius differences of cumulative row-fiber products. Generic packets are shared-factor convolutions of masked two-row Dirichlet-character sums, and their nonprincipal multiplicative energy is the connected fourth moment that remains to be bounded.

The row-diagonal spike shows why the order matters: a contribution may be power-saving after the complete scalar orbit sum and still saturate the raw fiber norm relative to its coefficient amplitude. Thus generic pruning cannot be removed inside a nonlinear norm merely by citing its scalar negligibility. This rules out a tempting shortcut and leaves a quantitatively specified gain target, $\omega > (y-j)/2$, for the sufficient character-moment route.

Accordingly, this paper advances the short–short branch by one positive closure and one exact reduction. It does not estimate the long connected character moment, control the large-divisor legs, prove the full TPC-18 residual-dispersion statement, breach sieve parity, or establish a fixed-shift Hardy–Littlewood asymptotic or a twin-prime lower bound.

References

[1] Sandro Bettin and Vorrapan Chandee. Trilinear forms with Kloosterman fractions. *Advances in Mathematics*, 328:1234–1262, 2018. doi: 10.1016/j.aim.2018.01.026. URL `https://doi.org/10.1016/j.aim.2018.01.026`.

[2] William Duke, John B. Friedlander, and Henryk Iwaniec. Bilinear forms with Kloosterman fractions. *Inventiones Mathematicae*, 128(1):23–43, 1997. doi: 10.1007/s002220050135. URL `https://doi.org/10.1007/s002220050135`.

[3] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI, 2004. doi: 10.1090/coll/053. URL `https://doi.org/10.1090/coll/053`.

[4] Alexandru Pascadi. Large sieve inequalities for exceptional Maass forms and the greatest prime factor of n^2+1. *Forum of Mathematics, Pi*, 14:e8, 2026. doi: 10.1017/fmp.2026.10025. URL `https://doi.org/10.1017/fmp.2026.10025`.

[5] Liang Wang. Two quadraticizations of a Möbius tail: LCM cutoff return, generic row determinants, and nonprimitive endpoint collapse. Preprint, repository commit `f418ea1a8921`, 2026.

[6] Liang Wang. Matched-spectrum gates in a primitive Möbius tail: Exact determinant–divisor normal forms, CRT–poisson duality, and a frequency-concentration gate. Preprint, repository commit `a6b64bf4fb50`, 2026.

[7] Liang Wang. Four-channel compression in a primitive Möbius tail: Centered divisor kernels, a single-leg completion wedge, and CRT-fiber discrepancy. Preprint, repository commit `b545ad5eb0db`, 2026.

[8] Thomas Wright. Trilinear Kloosterman fractions I: Partially fixed moduli and unbalanced convolutions, 2026. URL `https://arxiv.org/abs/2604.25177v1`. Preprint, arXiv:2604.25177v1, submitted 28 April 2026.